National Science Foundation
2415 Eisenhower Avenue
Alexandria, VA 22314

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Dear NSF SBIR Review Committee:

I write as Department Chair of Information Technology at Austin Community College, where I lead one of the largest IT programs in Texas. My department offers courses in printed circuit board design, electronics, and cybersecurity, including a PCB Design course (DFTG-2405) covering schematic capture, multilayer board layout, and electronic design automation. I am writing in support of ALTITUT LLC's VibeFab proposal led by PI Lu Sun.

Community colleges face a genuine instructional bottleneck in physical-computing courses. In our PCB Design course alone, students must navigate component selection, power budgets, packaging constraints, and manufacturing requirements. Each project is different. An instructor cannot individually review every design decision before parts are ordered and boards fabricated. The result is wasted materials, and repeated troubleshooting that does not scale.

The AI tools currently available do not resolve this. General-purpose language models generate plausible instructions but do not trace claims to validated references, identify missing requirements, or recognize when to escalate. Professional EDA tools like Cadence Allegro assume engineering fluency that community college students have not developed. Neither provides the evidence-bounded guidance that allows a learner to arrive at instructor review with complete requirements and fewer unsupported assumptions.

VibeFab is designed to occupy this gap. The system translates multimodal creative intent into traceable engineering requirements, adapts within machine-checkable reference-design boundaries, recommends decision-relevant physical tests, and abstains when evidence is insufficient. This is the correct model for AI-assisted hardware guidance where student safety and institutional accountability are non-negotiable.

If this proposal is awarded, Austin Community College will conduct a structured VibeFab pilot within our PCB Design and Career and Technical Education programs, provide ALTITUT LLC access to our faculty for deployment review, and document the experience for the broader community college network. Our institution is positioned to demonstrate that this technology can be deployed at scale.

I commend this proposal to your committee. VibeFab addresses a documented gap in tools available to community college students building physical projects, and PI Lu Sun and the ALTITUT LLC team have the focus to deliver.

Sincerely,

Fred Lover

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